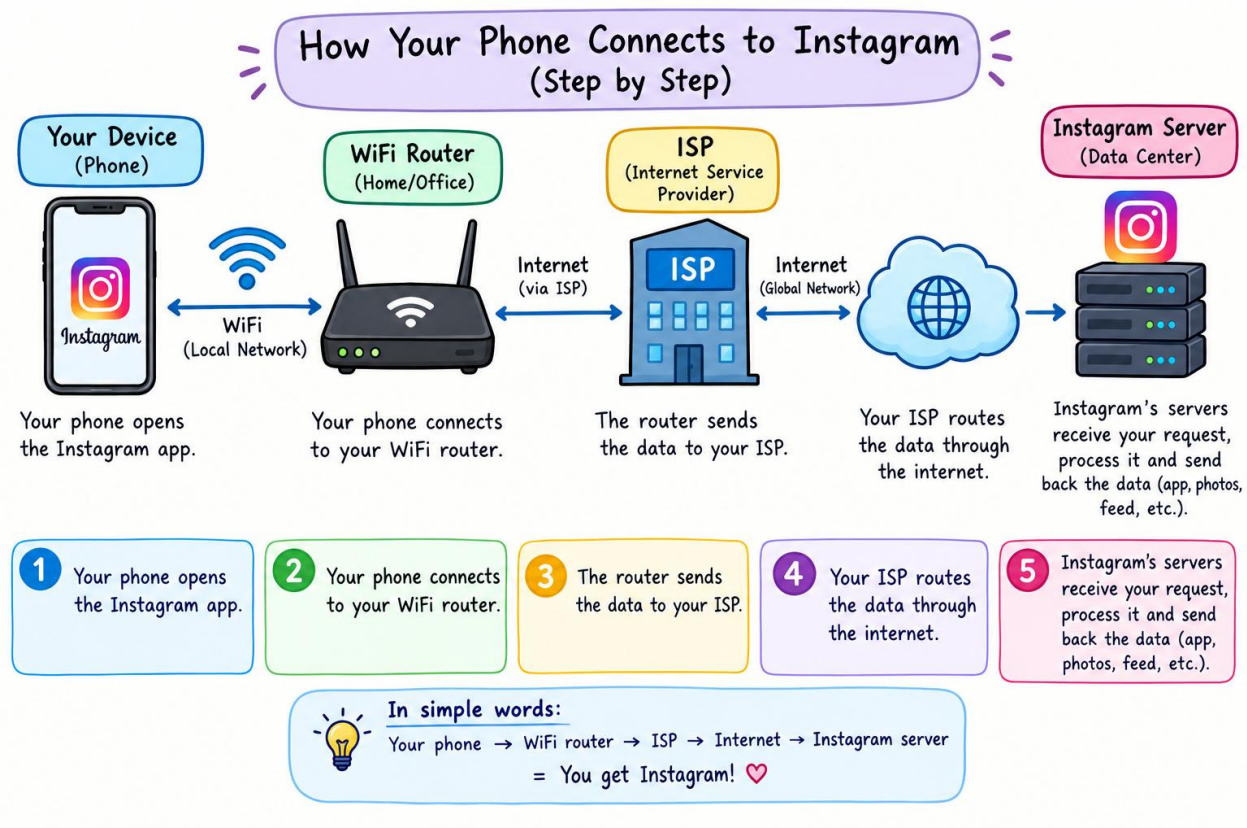
1.Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Ans.

Device	Network Type	Easy Explanation
1. **Smartphone**	PAN / WAN	Connects to nearby devices using Bluetooth (PAN) and to the internet using mobile data or Wi-Fi (WAN).
2. **Laptop**	LAN / WAN	Connects to a Wi-Fi network at home (LAN) and accesses the internet (WAN).
3. **Smart TV**	LAN / WAN	Connects to the home Wi-Fi (LAN) and uses the internet for Netflix, YouTube, etc. (WAN).
4. **Wi-Fi Router**	LAN / WAN	Connects devices inside the home (LAN) and connects them to the internet (WAN).
5. **Smartwatch**	PAN	Usually connects to your smartphone using Bluetooth, which is a Personal Area Network (PAN).

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Ans. **Difference between Internet and Intranet**

- **Internet:** It is a **public network** that anyone can use to access websites and online services from anywhere.
    - **Example:** When I use college/public Wi-Fi to open **Google, YouTube, or Instagram**, I am using the **Internet**.
- **Intranet:** It is a **private network** used only by members of an organization, such as a college or company.
    - **Example:** If my college Wi-Fi allows students to access an **internal college portal, attendance system, or notices**, that is an **Intranet**.

4.Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app. Mention the devices and networks involved, and what happens when you hit 'search'.

Ans. **WhatsApp Search**

1. When I search for something in **WhatsApp**, my smartphone sends the search request through **Wi-Fi/mobile data**.
2. The data goes from my phone to the **Wi-Fi router** and then through my **ISP** to the Internet.
3. The request reaches **WhatsApp's server**, where the search request is processed.
4. The server sends the required results/data back through the **Internet → ISP → router → my phone**.
5. Finally, the **search results appear on my WhatsApp screen**.